

5.5.3 Unusual parameters

Where unfamiliar parameters have been used to define the solution to a technical problem, the patentee is under a particular obligation to disclose all the information necessary to reliably determine the new parameter, in accordance with T.172/99, T.815/07, and T.593/09 (as recapitulated in T.1845/14).

In T.172/99 the board found that in the case of claimed subject-matter relying on a newly formulated and hence unfamiliar parameter to define the solution of a technical problem by which a relevant effect is achieved, the patentee is under a particular obligation to disclose all the information necessary reliably to define the new parameter not only (i) in a formally correct and complete manner such that its values can be obtained by a person skilled in the art without undue burden, but also (ii) in a manner which reliably retains the validity of the parameter for the solution of the technical problem for the application or patent in suit as a whole in the sense that the values routinely obtained will not be such that the claimed subject matter covers variants incapable of providing the relevant effect or, therefore, of solving the associated technical problem (followed in numerous decisions; see, for example, T.914/01, T.179/05, T.75/09 and T.1287/15 (dispute whether unfamiliar parameter (yes) – patentee's particular obligation identified in T.172/99 not met)).

The board in T.54/17 cited the settled case law that two requirements had to be met for the purposes of Art.100(b) EPC: the skilled person had to be able, firstly, to derive from the patent specification at least one way of carrying out the claimed invention and, secondly, to carry out the invention over the whole range claimed. The board held that the (second) requirement was particularly important, especially where a claim contained unusual parameters (parameters "operational result" and "reaction pattern" – certain meaning in German – not readily understood here as having a technical meaning). The board referred to the conditions listed in T.172/99. The board concluded that the skilled person was unable to carry out the invention.

The board in T.602/10 found that the proprietor had deliberately decided to use a method for determining rugosity which was different from the one commonly used in the state of the art. It was therefore its duty to provide full information with regard to the means and the procedures for implementing said method. In general terms, when the issue of sufficiency concerned the description of a method for determining a parameter, the less common the method the more accurate the information provided in the description should be. In the case at issue, in the absence of any known prior-art work applying the same method used in the patent for measuring the rugosity, the skilled person had to rely primarily on the teaching of the patent to put the method into practice.

In T.131/03 the board observed that, once an opponent had established a strong presumption that unusual parameters used to define the claimed subject-matter were inherently disclosed in the prior art, the patent proprietor could not simply claim the benefit of the doubt. It was instead incumbent on it to establish to what extent the parameters it had chosen to use to define its invention actually distinguished the claimed subject-matter from the prior art.

The board in T.1287/15 decided that the "incremental adsorption capacity" parameter was a newly formulated and hence unfamiliar parameter. This case was similar to T.172/99, but was not comparable with T.231/01, in which the relevant parameters had been sufficiently described by reference to their measuring methods. In T.1287/15 the principal issue was the understanding of the notion and not so much the availability of a measuring method, but considerations in the catchword of T.172/99 were nevertheless relevant. The patent proprietor (appellant) was under a particular obligation to disclose all information necessary to reliably define the new parameter it had chosen to introduce although an established parameter was available. That particular obligation, identified in T.172/99, had not been met.

In T.1414/08 the board observed that sufficiency of disclosure might be questionable if specific values of an unusual parameter are formulated in a patent as essential to the invention but no method of measuring that parameter is either known in the art or disclosed in the patent. But in the case at hand the parameter (tensile strength) was not unusual. Depending on the method of measurement, there exists an uncertainty as to the actual end values of the range for the tensile strength mentioned in the independent claims (issue of Art. 84 EPC rather than of Art. 83 EPC).

In T.1553/16 the board held that the issue at stake was not related to an alleged ambiguity in the determination of a parameter, which in some cases could indeed be a matter of clarity, but rather to a lack of essential information needed to run a specific procedure (namely the APTF-2 procedure), which had not been shown to be usual in the art but was necessary to determine an unusual feature (F time) mentioned in the operative claims. In other words, the lack of information did not result in the claim being unduly broad or unclearly delimited, but derived from the presence of a parameter which was in itself very specific, but whose method of measurement had been **kept secret**.

See also T.1452/16 (unusual parameters – alleged prior use – public availability of a product): addressing the fact that it had been possible for the patent proprietor to obtain and test samples, the board found that, when samples manufactured after the priority date were tested, it was legitimate to ask whether the results obtained were representative of the ratios present in the prior-art samples. However, where a patent proprietor used a parameter not used in the prior art, it bore the burden of proving that the prior art did not fall within the terms of the claim.

Other decisions: T.1764/06 (no benefit of the doubt – see catchword); T.288/06 (unusual method of measuring parameter); T.815/07; T.1920/09 (unusual parameter and presumption); T.484/05; T.1995/15 (implicit feature – where an unusual parameter is used, onus is on the applicant to establish novelty over prior art if there no reason to doubt that the prior art implicitly meets the parameter); T.615/19 (decisions establishing insufficient disclosure owing to unclear or inadequate information on the method of determining an unusual parameter held inapplicable to case in hand because the skilled person would not have encountered any difficulty).

Cf. T.1900/17 (fabric conditioners – use of a calculated partition coefficient such as ClogP for selecting nonionics – thorough discussion of whether ClogP was an unusual parameter).

6. Reproducibility

6.1. Repeatability

In T.281/86 (OJ 1989, 202), it was held that there is no requirement under Art. 83 EPC 1973 according to which a specifically described example of a process must be exactly repeatable. Variations in the constitution of an agent used in a process are immaterial to the sufficiency of the disclosure provided the claimed process reliably leads to the desired products. See also T.292/85 (OJ 1989, 275); T.299/86 (OJ 1988, 88); T.181/87, T.212/88 (OJ 1992, 28); T.182/89 (OJ 1991, 391) and T.19/90 (OJ 1990, 476).

In G.1/03 (point 2.5 of the Reasons) the Enlarged Board of Appeal indicated that a lack of reproducibility of the claimed invention is relevant under the requirements of sufficiency of disclosure if the technical effect is a technical feature of the claim, since then it is a feature characterising the subject-matter claimed (T.1079/08). A lack of reproducibility of the claimed invention (i.e. a failure of the claimed features to deliver the effect aimed for) is seen to represent, in the case of an effect which is not expressed in a claim but is part of the problem to be solved, "a problem of inventive step". If an effect is expressed in a claim, there is lack of sufficient disclosure (G.1/03, OJ, 2004, 413, and T.939/92, OJ 1996, 309, cited by T.2001/12; and more recently in T.1845/14).

According to established case law, the requirement of sufficiency of disclosure of Art. 83 EPC is not met if an effect expressed in the claim cannot be reproduced (summary of the case law in T.1473/13; see also T.1845/14).

In T.161/18 the application used an artificial neural network to transform the blood pressure curve measured on the periphery into the equivalent aortic pressure. As regards how the **neural network** according to the invention was trained, the application disclosed only that the input data should cover a wide range of patients differing in age, sex, constitution type, state of health, etc. to prevent the network from becoming specialised. However, it did not disclose what input data were suitable for training the network or even a suitable set of data for solving the technical problem in question. As a result, the skilled person could not reproduce the network's training and so could not carry out the invention. The invention, which was based on automated learning, in particular in connection with an artificial neural network, was thus insufficiently disclosed, because the training it involved could not be reproduced owing to a lack of disclosure in this regard.

6.2. Hypothetical embodiments

In T.515/00 the board pointed out that an invention cannot be considered to be irreproducible merely because a claim encompasses a hypothetical embodiment which lies outside the breadth of the claim as determined by the Protocol on the Interpretation of Art. 69 EPC 1973, which embodiment cannot be reproduced (endorsed in T.519/07).

6.3. Variants

If the only embodiment disclosed with concrete details in a patent is not disclosed in a manner sufficiently complete for the claimed invention to be carried out by a person skilled in the art on the date of priority with respect to the fundamental scope of said invention, it is of no significance with regard to the question of sufficient disclosure whether on the relevant date of filing a variant could have been carried out if the variant, although it is covered by the wording of the patent claim, does not fall within the fundamental scope of the claimed invention with regard to the teaching of the patent due to a lack of comparable technical success (T. 1173/00, OJ 2004, 16).

The board went on to state that if an invention is insufficiently disclosed, it is of no relevance whether it was objectively impossible to provide the **missing information** on the date of priority. The decisive issue is whether the invention is disclosed in a manner sufficiently complete for it to be carried out by an average person skilled in the art on the date of priority, with knowledge of the patent and on the basis of that person's common general knowledge.

6.4. Use to which invention is put

Where a disadvantage of an invention (in this case the risk of injury to users) could prevent its use, this is not an obstacle to reproducibility provided that the otherwise desired result is achieved by the technical teaching disclosed in the patent in suit (T. 881/95, see also T. 468/09). An Art. 83 EPC objection concerning the absence of any detailed indication of the **use** envisaged for the products cannot succeed, as Art. 83 EPC merely requires the invention to be sufficiently disclosed (see e.g. T. 866/00).

6.5. Reach-through claims

In T. 1063/06 (OJ 2009, 516) the board held that a formulation of a claim whereby functionally defined chemical compounds were to be found by means of a new kind of research tool using a screening method set out in the description constituted a reach-through claim which was also directed to future inventions based on the one now being disclosed. The applicant was entitled to claim patent protection only for his actual contribution to the art and not to reserve an unexplored field of research.

6.6. Reproducibility without undue burden

The disclosure must be reproducible without undue burden.

6.6.1 Occasional failure

It suffices for the disclosure of an invention that the means intended to carry out the invention are clearly disclosed in technical terms which render them implementable and that the intended result is achieved at least in some, equally realistic, cases (T. 487/91). The occasional failure of a process as claimed does not impair its reproducibility if only a few attempts are required to transform failure into success, provided that these attempts

are kept within reasonable bounds and do not require an inventive step (T.931/91). The skilled person is used to occasional failures when testing a technical teaching (T.14/83, cited in T.1133/08).

In ex parte case T.1111/14 (human hepatocytes), the board, disagreeing with the department of first instance's finding on Art. 83 EPC (infringed), cited the above finding from T.931/91 and added that, as a matter of fact, occasional failure was the rule rather than the exception in the technical field at issue.

A claim is an attempt to define a device in terms of ideal conditions, i.e. those required for its theoretically optimal or nominal operation. However, when considering a claim, the skilled person will readily understand that the conditions of actual operation will not be the ideal ones defined there. In T.383/14 (sorting table for grape harvest), the board found that, on reading the claim at issue, the skilled person would immediately grasp how the table would operate **in practice** after a harvest and so understand its terms in a sense compatible with the actual operation of all mechanical devices, whose reliability or success rate was always less than 100% and even lower in the specific case of sorting or grading.

The board in T.38/11 summarised the case law for making a case of insufficiency of disclosure (identifying gaps in information), and in the case at issue stated that the appellant (patentee) itself argued that a synergistic effect of a composition depended on a range of parameters and was rather an exceptional situation. As such parameters were not disclosed, it followed that the patent did not suffer from an occasional failure, but from a lack of a concept fit for generalisation. The situation may be aptly denoted as an invitation to carry out a research programme, based on trial and error, with limited chances of success (see T.435/91 (OJ 1995, 188) and T.809/07). In accordance sufficiency of disclosure could not be acknowledged.

6.6.2 Routine selection

Reproducibility is not impaired if the selection of the values for various parameters is a matter of routine and/or if further information is supplied by examples in the description (T.107/91).

The board in T.764/14 concluded that the skilled person was able, based on common general knowledge and corresponding routine variation of experimental conditions, to complement the information contained in paragraph [0031] of the patent in suit and, thus, to determine (possibly with some slight uncertainty but) without undue burden the surface Na baseline value for a given carrier.

See also chapter II.C.6.6.8 "Calibration and identifiable measurement method".

6.6.3 Wrong citations

Wrongly citing a method of measuring an essential product parameter may constitute insufficient disclosure (T.1250/01, applied in T.484/05 (unusual parameter)). In T.1250/01 the consequence of an error in the description was that no method for measuring an